Java Collections & Stream Programs

Answers – Numbered Programs

Answer 1

```
import java.util.*;
class Main {
    public static void main(String[] args) {
         List<Integer> list=
                Arrays.asList(10, 20, 30, 20, 40, 10, 50);
                Set<Integer> set=new HashSet<>();
                Set<Integer> dup=new LinkedHashSet<>();
                for(int n:list){
                    if(set.contains(n)){
                        dup.add(n);
                    }
                    else{
                        set.add(n);
                    }
                }
                System.out.print(dup);
    }
}
```

Answer 2

```
import java.util.*;
class Main {
    public static void main(String[] args) {
         List<Integer> list =
                Arrays.asList(10, 20, 10, 30, 20, 10);
                Map<Integer,Integer> map=new LinkedHashMap<>();
                for(int n:list){
                    map.put(n,map.getOrDefault(n,0)+1);
                }
                for(Map.Entry<Integer,Integer> e:map.entrySet()){
                System.out.println(e.getKey()+" = "+e.getValue());

                }
    }
}
```

Answer 3

```
import java.util.*;
class Main {
    public static void main(String[] args) {
         List<Integer> list=
                Arrays.asList(10, 50, 20, 80, 40, 70);
               int first=Integer.MIN_VALUE;
               int second=Integer.MAX_VALUE;
               for(int i:list){
                   if(i>first){
                       second=first;
                       first=i;
                   }
                   else if(i>second&& i!=first){
                       second=i;
                   }
               }
                System.out.print(second);
    }
}
```

Answer 4

```
import java.util.*;
class Main {
    public static void main(String[] args) {
              List<Integer> list1 =
                Arrays.asList(10, 20, 30, 40, 50);
```

```
        List<Integer> list2 =
                Arrays.asList(30, 40, 60, 70, 80);
        for(int n:list1){
            if(list2.contains(n)){
                System.out.print(n+" ");
            }
        }
    }
}
```

Answer 5

```
import java.util.*;

class Main {
    public static void main(String[] args) {

        List<Integer> list =
                Arrays.asList(10, 20, 10, 30, 20, 40, 30, 50);

        Map<Integer, Integer> map = new LinkedHashMap<>();

        for (int n : list) {
            map.put(n, map.getOrDefault(n, 0) + 1);
        }

        for (Map.Entry<Integer, Integer> e : map.entrySet()) {

            if (e.getValue() == 1)
                System.out.println(e.getKey());

            break;
        }
    }
}
```

Answer 6

```
import java.util.*;
class Main {
    public static void main(String[] args) {
      List<Integer> list =
                Arrays.asList(10, 20, 30, 20, 40, 10);
                 Set<Integer> set=new LinkedHashSet<>();
                 for(int n:list){
                     if(set.contains(n)){
                         System.out.print(n);
                         break;
                     }
                     else{
                         set.add(n);
                     }
                 }
    }
}
```

Answer 7

```
import java.util.*;

class Employee{
    int id;
    String name;
    double salary;
    Employee(int id,String name,double salary){
        this.id=id;
        this.name=name;
        this.salary=salary;
    }
    public int getId(){
        return id;
    }
    public String getName(){
        return name;
    }
    public double getSalary(){
```

```
        return salary;
    }
}

class Main {
    public static void main(String[] args) {
      List<Employee> list=Arrays.asList(
          new Employee(1,"Raghu",75000.00),
      new Employee(2,"Naik",35000.00),
      new Employee(3,"Kurra",45000.00)
      );

      list.sort(Comparator.comparing(Employee::getSalary));

     // list.sort(Comparator.comparing(Employee::getSalary).reversed());


      for(Employee e:list){
          System.out.println(e.getId()+" "+e.getName());
      }
    }
}
```

Answer 8

```
import java.util.*;

class Employee{
    int id;
    String name;
    double salary;
    Employee(int id,String name,double salary){
        this.id=id;
        this.name=name;
        this.salary=salary;
    }
    public int getId(){
        return id;
    }
    public String getName(){
        return name;
    }
    public double getSalary(){
        return salary;
    }
}
class Main {
    public static void main(String[] args) {
      List<Employee> list=Arrays.asList(
          new Employee(1,"Raghu",75000.00),
      new Employee(2,"Naik",35000.00),
      new Employee(3,"Kurra",45000.00),
      new Employee(4,"Ram",45000.00)
      );

      list.sort(Comparator.comparing(Employee::getSalary)
      .thenComparing(Employee::getName));



      for(Employee e:list){
          System.out.println(e.getId()+" "+e.getName());
      }
    }
}
```

Answer 9

```
import java.util.*;

public class CharacterFrequency {

    public static void main(String[] args) {

        List<String> list = Arrays.asList("programming");

        Map<Character, Integer> frequency =
                new LinkedHashMap<>();
```

```
                //String str = list.get(0);

        for (String str : list) {

            for (char ch : str.toCharArray()) {

                frequency.put(
                    ch,
                    frequency.getOrDefault(ch, 0) + 1
                );
            }
        }

        for (Map.Entry<Character, Integer> entry
                : frequency.entrySet()) {

            System.out.println(
                entry.getKey() + " -> " + entry.getValue()
            );
        }
    }
}
```

Answer 10

```
import java.util.*;

class Employee {
    int id;
    String name;
    double salary;

    Employee(int id, String name, double salary) {
        this.id = id;
        this.name = name;
        this.salary = salary;
    }
}
class Main {

    public static void main(String[] args) {

        List<Employee> employees = Arrays.asList(
            new Employee(101, "Ravi", 50000.0),
            new Employee(102, "Anil", 75000.0),
            new Employee(103, "Suresh", 60000.0),
            new Employee(104, "Raj", 45000.0)
        );

       Employee highest = employees.stream()
        .max(Comparator.comparing(e -> e.salary)).orElseThrow();


System.out.println(
    highest.name + " = " + highest.salary
);
    }
}
```

Answer 11

```
import java.util.*;
import java.util.stream.*;

public class EvenNumbersStream {

    public static void main(String[] args) {

        List<Integer> numbers =
                Arrays.asList(10, 15, 20, 25, 30, 35, 40);

        List<Integer> evenNumbers = numbers.stream()
                .filter(n -> n % 2 == 0)
                .toList();

        System.out.println("Even numbers: " + evenNumbers);
    }
}
```

Answer 12

```
import java.util.*;
import java.util.stream.*;

public class EvenNumbersStream {

    public static void main(String[] args) {

          List<Integer> numbers =
                Arrays.asList(10, 20, 10, 30, 20, 40, 30);

        List<Integer> unique = numbers.stream()
                .distinct()
                .toList();

        System.out.println(unique);
    }
}
```

Answer 13

```
import java.util.*;
import java.util.stream.*;

 class Main {

    public static void main(String[] args) {

        List<Integer> numbers =
                Arrays.asList(10, 20, 30, 20, 40, 10, 50, 30);

        Set<Integer> seen = new HashSet<>();

        List<Integer> duplicates = numbers.stream()
                .filter(n -> !seen.add(n))
                .toList();

        System.out.println(
                "Duplicate elements: " + duplicates);
    }
}
```

Answer 14

```
import java.util.*;

public class Main {

    public static void main(String[] args) {

        List<Integer> numbers =
                Arrays.asList(10, 50, 20, 80, 40, 70);

        int secondHighest = numbers.stream()
                .sorted(Comparator.reverseOrder())
                .skip(1)
                .findFirst()
                .get();

        System.out.println(
                "Second highest: " + secondHighest);
    }
}
```

Answer 15

```
import java.util.*;

public class Main {

    public static void main(String[] args) {

        List<Integer> numbers =
```

```
                Arrays.asList(50, 10, 30, 20, 40, 60);

        int secondLowest = numbers.stream()
                .sorted()
                .skip(1)
                .findFirst()
                .get();

        System.out.println(
                "Second lowest: " + secondLowest);
    }
}
```

Answer 16

```
import java.util.*;

class Employee {

    int id;
    String name;
    double salary;

    Employee(int id, String name, double salary) {
        this.id = id;
        this.name = name;
        this.salary = salary;
    }
}

public class Main {

    public static void main(String[] args) {

        List<Employee> employees = Arrays.asList(
                new Employee(101, "Ravi", 50000),
                new Employee(102, "Anil", 80000),
                new Employee(103, "Suresh", 60000),
                new Employee(104, "Raj", 70000)
        );

        Employee highest = employees.stream()
                .max(Comparator.comparingDouble(e -> e.salary))
                .get();

        System.out.println(
                "Highest salary employee: "
                + highest.name + " = "
                + highest.salary);
    }
}
```

Answer 17

```
import java.util.*;

class Employee {

    int id;
    String name;
    double salary;

    Employee(int id, String name, double salary) {
        this.id = id;
        this.name = name;
        this.salary = salary;
    }
}

public class Main {

    public static void main(String[] args) {

        List<Employee> employees = Arrays.asList(
                new Employee(101, "Ravi", 50000),
                new Employee(102, "Anil", 80000),
                new Employee(103, "Suresh", 60000),
                new Employee(104, "Raj", 70000)
```

```
        );

        Employee secondHighest = employees.stream()
                .sorted(Comparator.comparingDouble(
                        (Employee e) -> e.salary
                ).reversed())
                .skip(1)
                .findFirst()
                .get();

        System.out.println(
                "Second highest salary employee: "
                + secondHighest.name + " → "
                + secondHighest.salary);
    }
}
```

Answer 18

```
import java.util.*;
import java.util.stream.*;

class Employee {

    int id;
    String name;
    double salary;
    String department;

    Employee(int id, String name, double salary, String department) {
        this.id = id;
        this.name = name;
        this.salary = salary;
        this.department = department;
    }
}

public class Main {

    public static void main(String[] args) {

        List<Employee> employees = Arrays.asList(
            new Employee(101, "Ravi", 50000, "IT"),
            new Employee(102, "Anil", 60000, "HR"),
            new Employee(103, "Suresh", 70000, "IT"),
            new Employee(104, "Raj", 40000, "HR"),
            new Employee(105, "Kumar", 55000, "Sales")
        );

        Map<String, List<Employee>> grouped =
            employees.stream()
                .collect(Collectors.groupingBy(
                    e -> e.department
                ));

        for (Map.Entry<String, List<Employee>> entry : grouped.entrySet()) {

            System.out.print(entry.getKey() + ": ");

            entry.getValue().forEach(
                e -> System.out.print(e.name + " ")
            );

            System.out.println();
        }
    }
}
```

Answer 19

```
import java.util.*;
import java.util.stream.*;

class Employee {

    int id;
    String name;
    double salary;
```

```
    String department;

    Employee(int id, String name, double salary, String department) {
        this.id = id;
        this.name = name;
        this.salary = salary;
        this.department = department;
    }
}

public class Main {

    public static void main(String[] args) {

        List<Employee> employees = Arrays.asList(
            new Employee(101, "Ravi", 50000, "IT"),
            new Employee(102, "Anil", 60000, "HR"),
            new Employee(103, "Suresh", 70000, "IT"),
            new Employee(104, "Raj", 40000, "HR"),
            new Employee(105, "Kumar", 50000, "Sales")
        );

        Map<String, Double> averageSalary =
            employees.stream()
                .collect(Collectors.groupingBy(
                    e -> e.department,
                    Collectors.averagingDouble(
                        e -> e.salary
                    )
                ));

        averageSalary.forEach(
            (department, average) ->
                System.out.println(
                    department + " → " + average
                )
        );
    }
}
```

Answer 20

```
import java.util.*;
import java.util.stream.*;

class Employee {

    int id;
    String name;
    double salary;
    String department;

    Employee(int id, String name, double salary, String department) {
        this.id = id;
        this.name = name;
        this.salary = salary;
        this.department = department;
    }
}

public class HighestPaidByDepartment {

    public static void main(String[] args) {

        List<Employee> employees = Arrays.asList(
            new Employee(101, "Ravi", 50000, "IT"),
            new Employee(102, "Anil", 60000, "HR"),
            new Employee(103, "Suresh", 70000, "IT"),
            new Employee(104, "Raj", 40000, "HR"),
            new Employee(105, "Kumar", 55000, "Sales"),
            new Employee(106, "Arjun", 65000, "HR")
        );

        Map<String, Optional<Employee>> result =
            employees.stream()
                .collect(Collectors.groupingBy(
                    e -> e.department,
                    Collectors.maxBy(
                        Comparator.comparingDouble(
                            e -> e.salary
```

```
                        )
                    )
                ));

        result.forEach((department, employee) -> {

            Employee e = employee.get();

            System.out.println(
                department + " → "
                + e.name + " → "
                + e.salary
            );
        });
    }
}
```